Customer Sentiment Analysis & Reporting for E-commerce

Capstone project · an end-to-end ML system that turns raw product reviews into live sentiment predictions and business dashboards

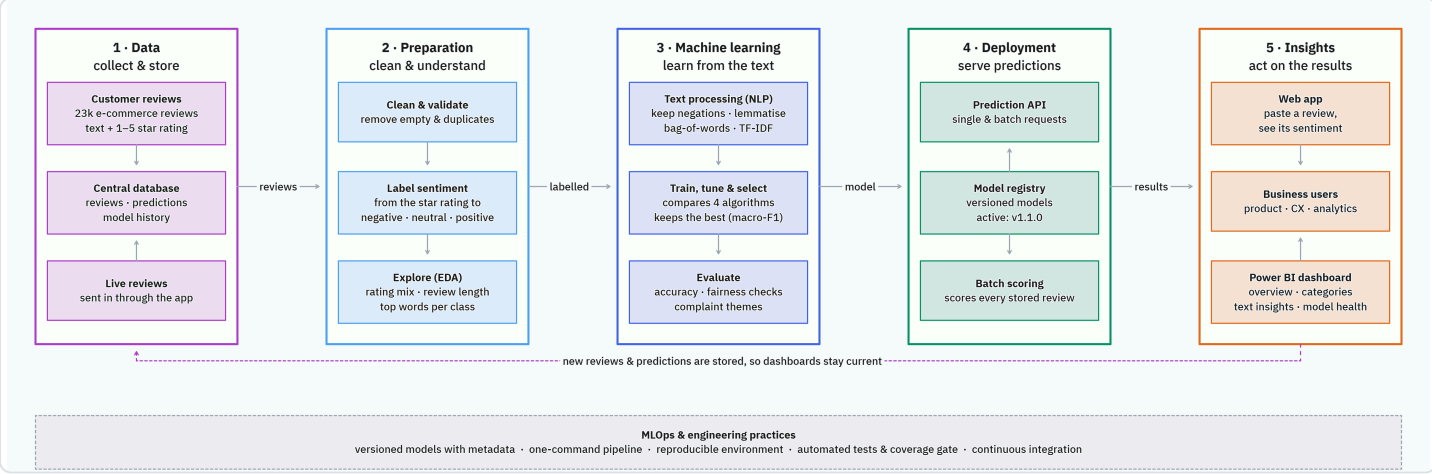
Model v1.1.0 · Python 3.11 · scikit-learn · NLTK · SQLite · Flask · Power BI

PROBLEM & GOAL

Star ratings say *how* happy customers are, not *why*, and new feedback (support messages, social posts) often has no rating at all. The goal is to classify review text as **negative**, **neutral** or **positive**, store every prediction, serve the model through an API, and show the results to business teams.

DATA

- **Women's E-Commerce Clothing Reviews** (public): 23,486 reviews with title, text, 1–5 stars, department and class.
- Cleaning removed 845 empty reviews and 1 duplicate → **22,640** reviews.
- Labels from stars: 1–2 negative, 3 neutral, 4–5 positive. Imbalanced: 77.1% / 12.5% / 10.5%.
- Stratified split 70 / 15 / 15 (train / validation / test). The test set is used once, for the final score.



MACHINE LEARNING

- **Text processing:** lowercase; strip URLs, HTML and digits; tokenise; remove stop words but **keep negations** ("didn't", "dont" → "not"); lemmatise.
- **Features:** bag-of-words and TF-IDF over 1–2 word n-grams.
- **Models compared:** Naive Bayes (baseline), Logistic Regression, Random Forest, HistGradientBoosting.
- **Tuning:** Grid and randomized search with stratified k-fold CV. Imbalance handled with balanced class weights and a macro-F1 target.
- **Selection:** best validation macro-F1 wins; one pipeline object holds cleaning, features and the classifier, so training and serving can't drift.
- **Extra analysis:** K-Means clustering finds complaint themes (sizing, fabric quality, fit).

RESULTS (HELD-OUT TEST SET)

0.660	82.9%	0.654	6 pts
macro-F1	accuracy	baseline	max gap

Class	Precision	Recall	F1
Negative	0.58	0.60	0.59
Neutral	0.43	0.50	0.46
Positive	0.94	0.91	0.93

Winner: tuned Multinomial Naive Bayes (α = 0.5). All acceptance gates pass: macro-F1 ≥ 0.65, beats the baseline, minority recall ≥ 0.30. No department or age band is more than 6 points from the overall score.

Limitations: neutral is the weakest class, because 83% of 3-star reviews mix praise and complaints. Very short or off-topic text can lean positive. Labels come from star ratings, so they are a proxy.

DEPLOYMENT & REPORTING

- **REST API** (Flask): `POST /predict`, `/predict/batch` (≤ 100), `GET /reviews`, `/stats`, `/health`, `/model/info`. Local latency p50 ≈ 24 ms, p95 ≈ 134 ms.
- **Web app** at `/`: paste reviews and see the label and class probabilities.
- **Web dashboard** at `/dashboard`: overview, categories, text insights and model monitoring, read live from SQL.
- **Batch scoring** stores a prediction for every review; every API call is stored too.
- **Database:** SQLite with constraints and 7 reporting views, exported to CSV for **Power BI** (4-page layout plus a setup guide).
- **Privacy:** logs never contain review text; fairness is checked by department and age.

MLOPS & ENGINEERING

- Reproducible environment with **uv** (locked dependencies, pinned Python); fixed random seed.
- Versioned models with metadata (data hash, seed, library version, metrics) and a registry with promote and rollback.
- One-command pipeline: `make all` = ingest → EDA → train → batch-score → export (about 4½ min).
- **47 automated tests**, 74% coverage (gate 70%), ruff lint, GitHub Actions CI on every push.

RUN IT

`make setup` → `make data` → `make train` → `make serve`, then open `localhost:5000` and `/dashboard`. On Windows use `.\make.ps1 <target>`. Dataset goes in the data folder as `reviews.csv`.

COURSE MODULES COVERED

ML & data science ecosystem · NumPy · Pandas · Matplotlib · Seaborn · Scikit-learn · supervised learning · feature engineering · Model evaluation · clustering · NLP · Naive Bayes · SQL · Python + SQL · Flask · Power BI · ensembles · tuning · ML pipelines · MLOps · **Capstone:** this project

More detail in the repository: README, requirements, data, ML pipeline, database, API, dashboard, MLOps, ethics and testing documents, plus DEMO.md.

github.com/sanathpedapudi24/ecommerce-sentiment-analysis